

# 半夜来，天明去

**Don't kill an idea — let AI survey it for you**

**Jingzong Zhang**

# Motivation

## Students vs. Senior Researchers

- **Ideas come from:** talks, chats, labmates
- **Senior researcher:** one sentence gut call — "promising / dead end"
- **Student:** no such intuition yet
- **Proper way:** one week of literature survey per idea
- **Reality:** one survey per week, most ideas die from "too lazy to check"

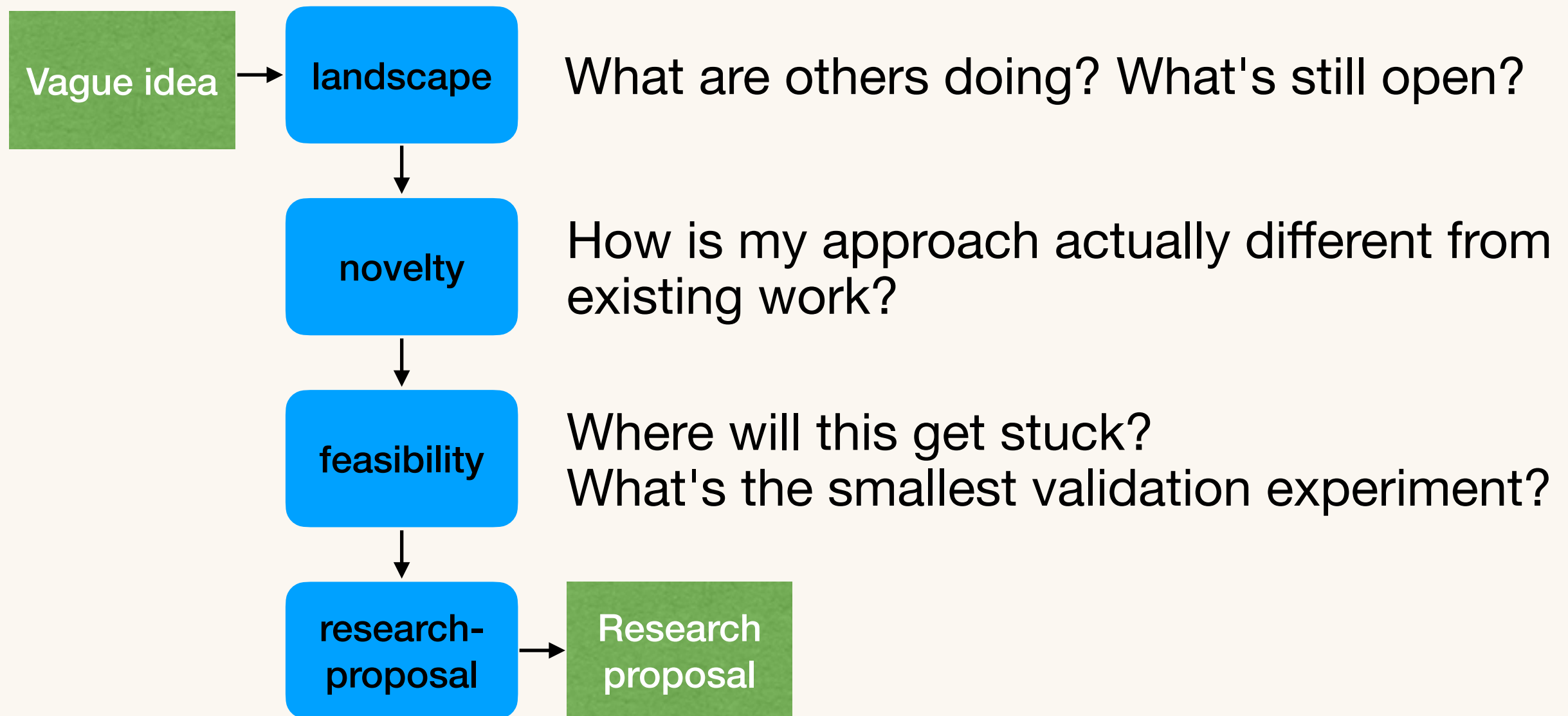
# Why Existing Tools Aren't Enough

## Off-the-shelf solutions don't match physics

|                      | Existing tools<br>(chatgpt deep research<br>/gpt-research...) | What physics research actually needs                                            |
|----------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Care about</b>    | Shipping engineering projects                                 | Whether derivations hold,<br>assumptions stand,<br>physical picture makes sense |
| <b>Equations</b>     | OCR errors from PDF parsing                                   | Every symbol must match the original                                            |
| <b>Figures</b>       | Treated as decoration                                         | Carefully designed by authors<br>must be read one by one                        |
| <b>Hallucination</b> | Model self-checks                                             | Source tracing + auditing                                                       |
| <b>Output</b>        | A short review article                                        | A research plan<br>with a full evidence chain                                   |

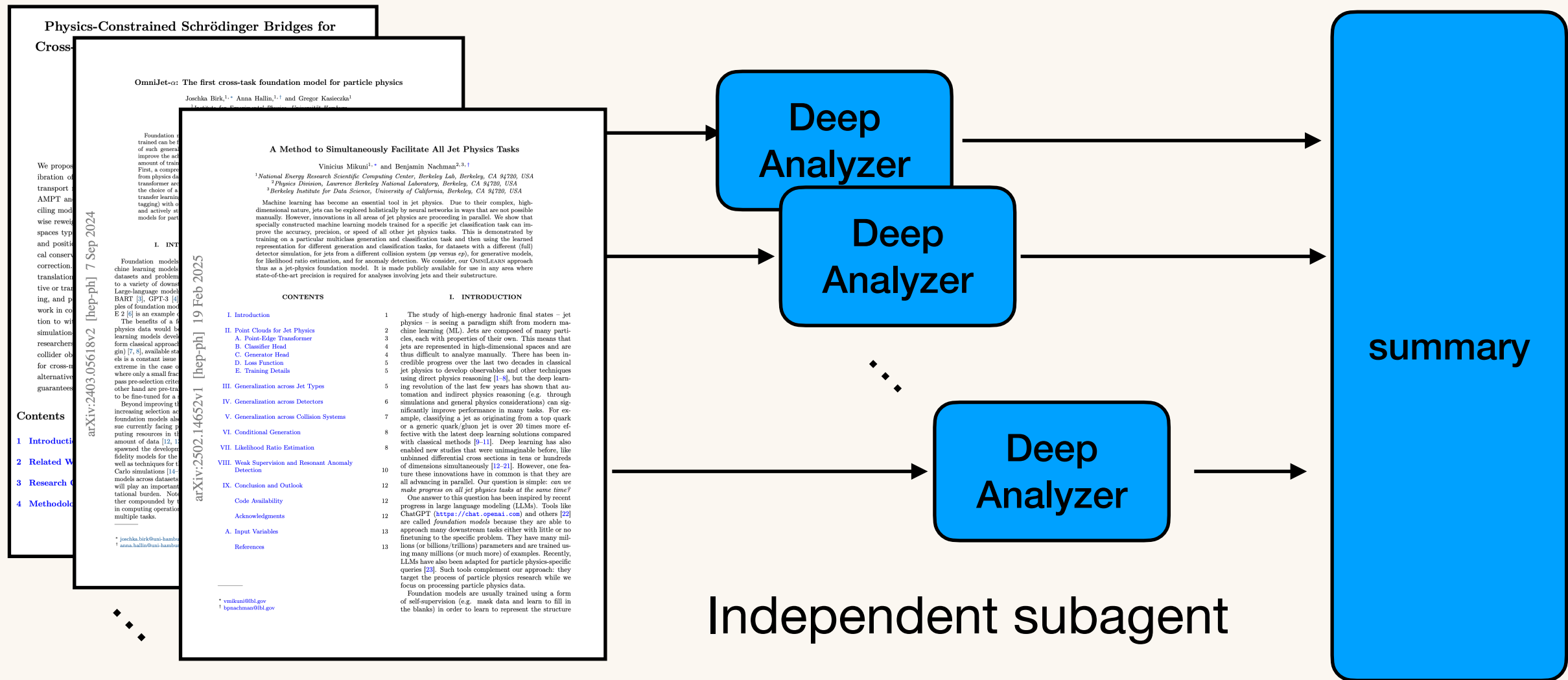
# Workflow

**One week of survey, one overnight run — 4 skills**



# Text Isolation

## Each paper read independently



# Final Output

## Physics-Constrained Schrödinger Bridges for Cross-Model Calibration in Heavy-Ion Collision Phenomenology

Jingzong Zhang

May 18, 2026

### Abstract

We propose a physics-constrained Schrödinger Bridge (SB) framework for cross-model calibration of heavy-ion collision phenomenology. The framework learns optimal stochastic transport maps between the final-state particle distributions of competing models such as AMPT and UrQMD without requiring event-level pairing. Existing approaches to reconciling model-dependent systematic differences—including discriminative regression and bin-wise reweighting—suffer from the curse of dimensionality in the  $10^3$ – $10^4$  dimensional phase spaces typical of heavy-ion events. They also lose correlated structures across momentum and position variables, and they provide no principled mechanism to enforce exact physical conservation laws such as energy-momentum or baryon-number conservation during the correction. We formulate cross-model alignment as an unpaired distribution-to-distribution translation problem, embedding hard physical constraints directly into the SB training objective or transport geometry. Manifold-aware drift parameterizations, constrained flow matching, and penalty terms preserve conservation laws along every diffusion trajectory. Recent work in collider physics demonstrates that diffusion models can learn momentum conservation to within 0.1% from data alone [1]. Neural optimal-transport calibration has closed simulation-data gaps in 128-dimensional latent spaces to discriminator AUC  $\approx 0.52$  [2], and researchers have demonstrated generative unfolding with Schrödinger Bridges on hadron-collider observables [3]. This proposal targets the first end-to-end probabilistic framework for cross-model alignment in heavy-ion physics. It provides a bidirectional, interpretable alternative to traditional reweighting with quantifiable transport costs and explicit physical guarantees.

### Contents

|   |                                    |    |
|---|------------------------------------|----|
| 1 | Introduction and Motivation        | 3  |
| 2 | Related Work and Literature Review | 5  |
| 3 | Research Objectives                | 10 |
| 4 | Methodology and Technical Approach | 12 |

## Rigor Audit Report

**Scope:** Scientific rigor only. No language/style edits unless required by a rigor check. **Date:** 2026-05-18

### Summary of Changes

| File                  | Change Type | Description                                                                                                                                                                                                                                                  |
|-----------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| abstract.tex          | None        | Passed all rigor checks.                                                                                                                                                                                                                                     |
| sec1_introduction.tex | Rigor fix   | Added missing <code>\label{sec:introduction}</code> .                                                                                                                                                                                                        |
| sec2_related_work.tex | Rigor fix   | Added missing <code>\label{sec:related-work}</code> ; restructured "Schrödinger Bridge theory and algorithms" subsection from paper-by-paper enumeration to thematic grouping (foundations, simulation-free algorithms, geometric extensions, applications). |
| sec3_objectives.tex   | Rigor fix   | Added missing <code>\label{sec:objectives}</code> .                                                                                                                                                                                                          |
| sec4_methodology.tex  | Rigor fix   | Added missing <code>\label{sec:methodology}</code> .                                                                                                                                                                                                         |
| sec5_innovation.tex   | Rigor fix   | Added missing <code>\label{sec:innovation}</code> .                                                                                                                                                                                                          |
| sec6_feasibility.tex  | Rigor fix   | Added missing <code>\label{sec:feasibility}</code> .                                                                                                                                                                                                         |
| sec7_timeline.tex     | Rigor fix   | Added missing <code>\label{sec:timeline}</code> .                                                                                                                                                                                                            |
| sec8_expected.tex     | Rigor fix   | Added missing <code>\label{sec:expected-results}</code> .                                                                                                                                                                                                    |

### Check 1 — Numerical-Claim Provenance

**Method:** For every specific number in prose, verify against (1) original TeX source via `tex_source_dir`, (2) `deep_analysis.md`, (3) `upstream_metadata.json` / `CONSISTENCY NOTES.md`.

## Language Audit Report

Generated: 2024-05-18T14:15:00Z

### Abstract — five-sentence formula

- Old:** "We propose a physics-constrained Schrödinger Bridge (SB) framework for cross-model calibration of heavy-ion collision phenomenology, learning optimal stochastic transport maps between the final-state particle distributions of competing models such as AMPT and UrQMD without requiring event-level pairing." **New:** "We propose a physics-constrained Schrödinger Bridge (SB) framework for cross-model calibration of heavy-ion collision phenomenology. The framework learns optimal stochastic transport maps between the final-state particle distributions of competing models such as AMPT and UrQMD without requiring event-level pairing." **Reason:** Split

# Equation Fidelity

TeX, not PDF

PDF

PDF

what the AI receives

OCR

$$\partial_{\mu} F_{\mu\nu} = j_{\nu}$$



$$\alpha_{\mu} F_{\mu\nu} = j_{\nu}$$

LaTeX:

tex

what the AI receives

$$\partial_{\mu} F_{\mu\nu} = j_{\nu}$$

$$\partial_{\mu} F_{\mu\nu} = j_{\nu}$$



# Forced close reading

## Figure-by-Figure, Equation-by-Equation, Table-by-Table

- Each figure → one image-reading tool call
- Each equation → same
- Each table → same
- Template declared up front — skipping any item fails the auto-check



# Traceable Evidence Chain

## Three lines of defense against hallucination

Report claims "accuracy = 0.92" or "significant improvement"

|

|— Level 1: Can I grep it in the cited paper's arXiv TeX source? ✓ pass

|— Level 2: Can I grep it in our deep-analysis notes? ✓ pass

|— Level 3: Can I grep it in the upstream stage report? ✓ pass

|— All three fail → rewrite as qualitative, or flag [needs fix]

# Human in the Loop

## Feedback section, read on the next run

```
## User Guidance Section
```

```
<!--
```

---

```
ARIS_GUIDANCE_START
```

```
The content below will be read on the next run as instructions.
```

```
Add corrections, additions, or new search directions.
```

```
Save the file and re-run /skill:idea-landscape to apply.
```

---

```
-->
```

```
- For deeper exploration (e.g. combining BranchSBM with HIC datasets), add detailed guidance below.
```

```
- I think you should survey more AI conference papers...
```

```
<!--
```

```
ARIS_GUIDANCE_END
```

---

```
-->
```

# A Real Example



Welcome to Kimi Code CLI!  
Send /help for help information.

Directory: ~/SB  
Session: 9293dfe5-8986-4e88-8804-97e8bd5b3afe  
Model: Kimi-k2.6

Tip: Spot a bug or have feedback? Type /feedback right in this session – every report makes Kimi better.

✨ /skill:research-proposal "使用Schrödinger Bridge方法解决重离子碰撞中跨唯象模型（AMPT, UrQMD, 实验数据）的模型依赖性问题，实现不同模拟器间末态粒子分布的对齐映射。" – literature-dir: idea-survey/ – output: proposal/ – author: "Jingzong Zhang" – language: en

- 用户要求我使用 **research-proposal** 技能来生成一个 *LaTeX* 研究提案。主题是通过 *Schrödinger Bridge* 方法解决重离子碰撞中跨唯象模型（AMPT... UrQMD, 实验数据）的模型依赖性问题，实现不同模拟器间末态粒子分布的对齐映射。

参数：

- 标题：使用 *Schrödinger Bridge*方法解决重离子碰撞中跨唯象模型的模型依赖性问题，实现不同模拟器间末态粒子分布的对齐映射
- *literature-dir: idea-survey/*
- *output: proposal/*
- *author: "Jingzong Zhang"*
- *language: en*

我需要按照技能文档的工作流逐步执行。让我先设置 *todo list*，然后开始 *Step 0*。

首先，我需要：

1. 解析参数
2. 创建目录
3. 验证 *idea-survey/* 目录存在
4. 检查共享参考文件是否存在

让我开始。

- Used **SetTodoList**
  - Step 0: Parse args, create dirs, verify inputs ←
  - Step 1: Extract upstream metadata & cross-check (Subagent A)
  - Step 2: Index literature assets (Subagent B)

# Limitations

## Slow + expensive

- Slow: each stage takes 2–3 hours
- Expensive: lots of tokens. I use Kimi (multimodal + cheap + smart enough). 199 RMB/month plan  $\approx$  2 ideas per week
- Good news: single idea is slow, but the deep-analysis library is shared — the next idea in the same direction only fills the gap, much faster
- Replace the human entirely? Not possible. That's why the feedback section exists.

**It doesn't replace physics intuition  
it lets more ideas seriously start  
the first mile.**